

Incident Investigation Report

Module 2: Endpoint Defense & Memory Analysis

Role: Senior SOC Tier 2 Analyst

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Date: 22 April 2026

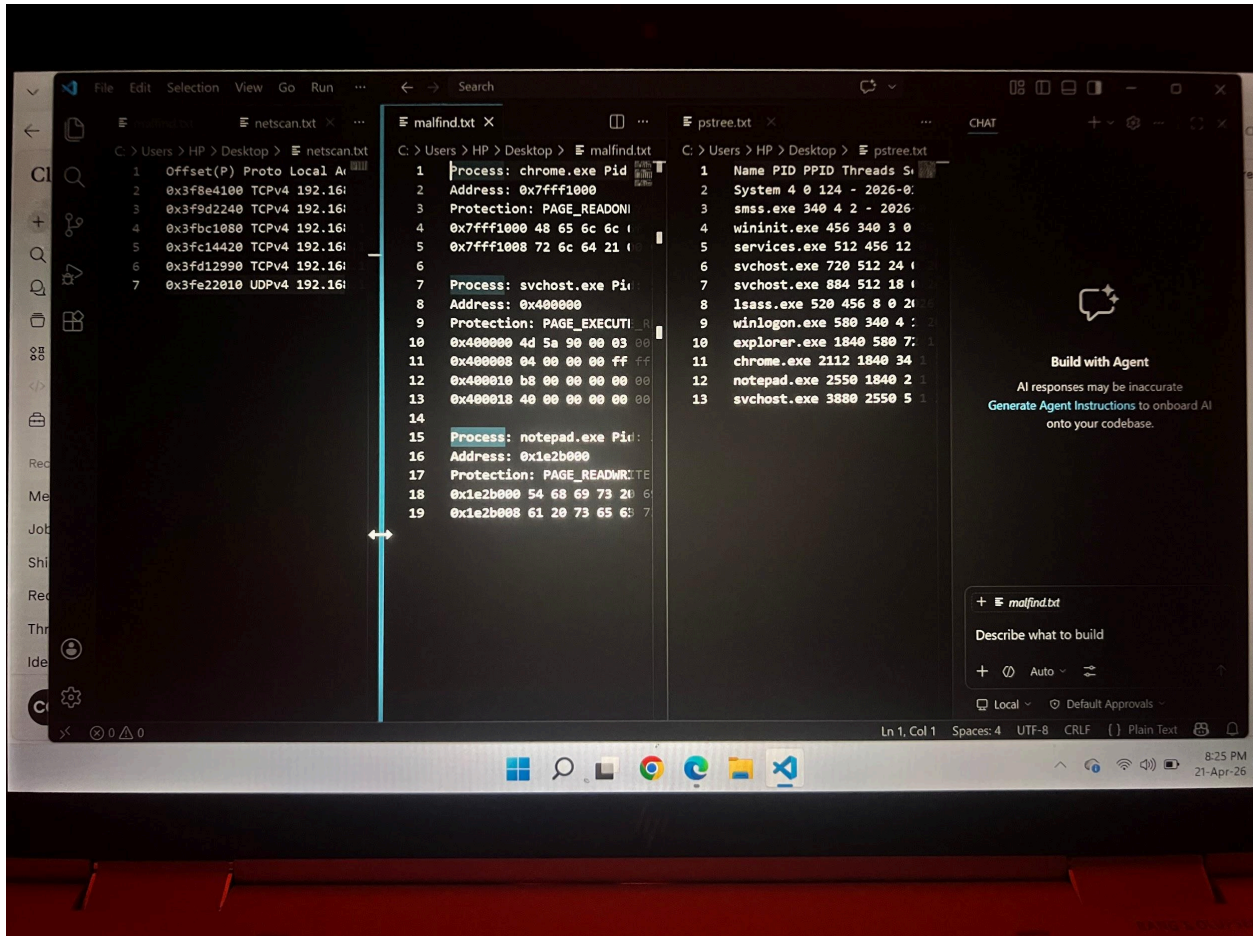
Executive Summary

This report documents a forensic investigation that was conducted on a workstation in the Finance Department after a trigger generated a Suspicious Network Connection alert.

The local antivirus failed to detect any malicious files on disk, prompting the capture of a full memory dump.

Three volatility output files (pstree_output.txt, malfind_results.txt, and netscan_results.txt) were analyzed carefully using VS Code Split View, the findstr command-line utility, CyberChef, and the MITRE ATT&CK Framework.

After the investigation, I successfully identified an active compromise that involved process injection into a legitimate Windows system process.



— Identified Compromised Process

After thorough examination and cross-referencing all three Volatility output files using the findstr command, the process identified as the malware vessel is svchost.exe with the Process ID (PID) 3880.

This was concluded after PID 3880 appeared consistently across all three output files (pstree_output.txt, malfind_results.txt, and netscan_results.txt) .

Unlike the other svchost running on the system, this particular process exhibited a behavior inconsistent with normal Windows operations exhibiting an anomalous parent-child relationship

with notepad.exe.

The screenshot shows a Windows desktop with a PowerShell terminal and several text files. The terminal displays the output of a 'findstr' command, which identifies 'notepad.exe' as the parent process of 'svchost.exe' (PID 3880). The text files show network connections, process details, and a process tree.

```
PS C:\Users\HP\Desktop> findstr "3880" pstree.txt malfind.txt netscan.txt
+ findstr "3880" pstree.txt malfind.txt netscan.txt
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (findstr:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\HP\Desktop> findstr "3880" pstree.txt malfind.txt netscan.txt
pstree.txt:svchost.exe 3880 2550 5 1 2026-03-10 14:46:12 malfind.txt:Process: svchost.exe Pid: 3880
netscan.txt:0x3fd12990 TCPv4 192.168.1.10:49552 185.112.55.20:443 ESTABLISHED 3880 svchost.exe
PS C:\Users\HP\Desktop>
```

— Parent-Child Analysis

The parent process of svchost.exe (PID 3880) was identified as notepad.exe with PID 2550.

This relationship is architecturally suspicious in a Windows environment because svchost.exe is a Windows system process that should only ever be prompted by services.exe (PID 512).

In this case, notepad.exe; a text editor application with no legitimate reason to launch system processes was used to spawn svchost.exe which is odd and a strong indicator of a malicious activity.

This abnormality between the parent-child relationship indicates that notepad.exe was exploited or engineered by the attacker as a launchpad for the attack.

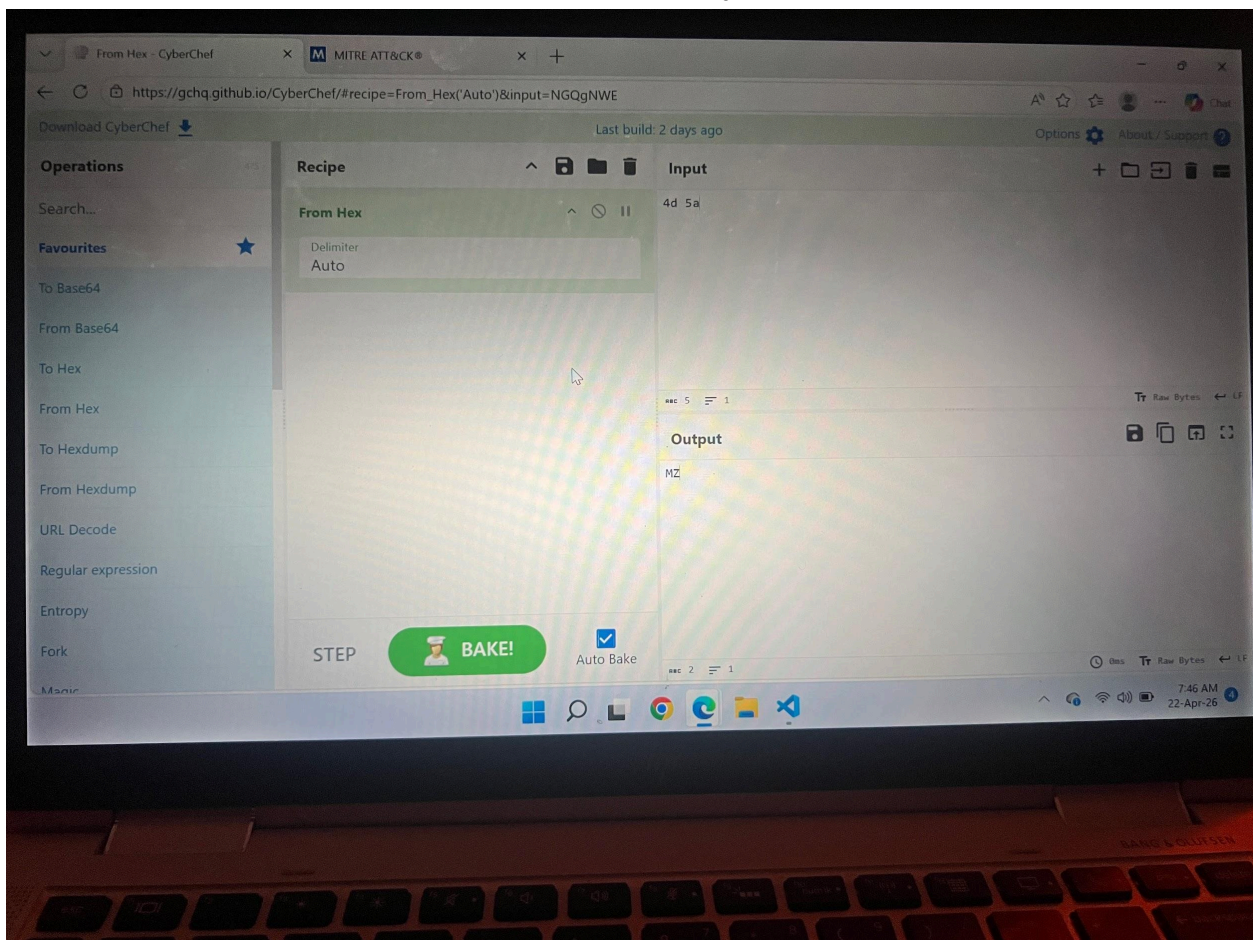
Memory Evidence

Analysis of the malfind_results.txt file revealed critical evidence of code injection within svchost.exe (PID 3880).

The memory region at address 0x400000 was flagged with a protection flag of PAGE_EXECUTE_READWRITE. While 0x400000 is a standard base address for loading executables, the READWRITE protection flags means that the memory can be read, written to or executed simultaneously, thereby violating standard security protocols that separates writable data from executable code.

Furthermore, the hexadecimal bytes 4d 5a were identified at the start of the memory region, which I confirmed using CyberChef to decode to the ASCII characters MZ (the standard Windows Portable Executable (PE) header) .

Therefore, the presence of an MZ header inside another process memory space is direct forensic evidence that a complete executable file was injected into svchost.exe.



C2 Identification

The analysis of the netscan_results.txt file identified an active outbound connection originating from svchost.exe (PID 3880).

This process was found communicating with the external IP address 185.112.55.20 over port 443.

The connection state was recorded as ESTABLISHED which confirmed that an active communication took place between the suspicious process and the remote server at the time the memory was captured.

The attacker deliberately chose port 443 (normally used for legitimate HTTPS web traffic) to hide the malicious communication because most firewalls allow port 443 traffic without questioning it.

Attack Classification

Based on the combined evidence gathered from all three Volatility output files, this attack is classified as Process Hollowing (MITRE ATT&CK T1055.012)

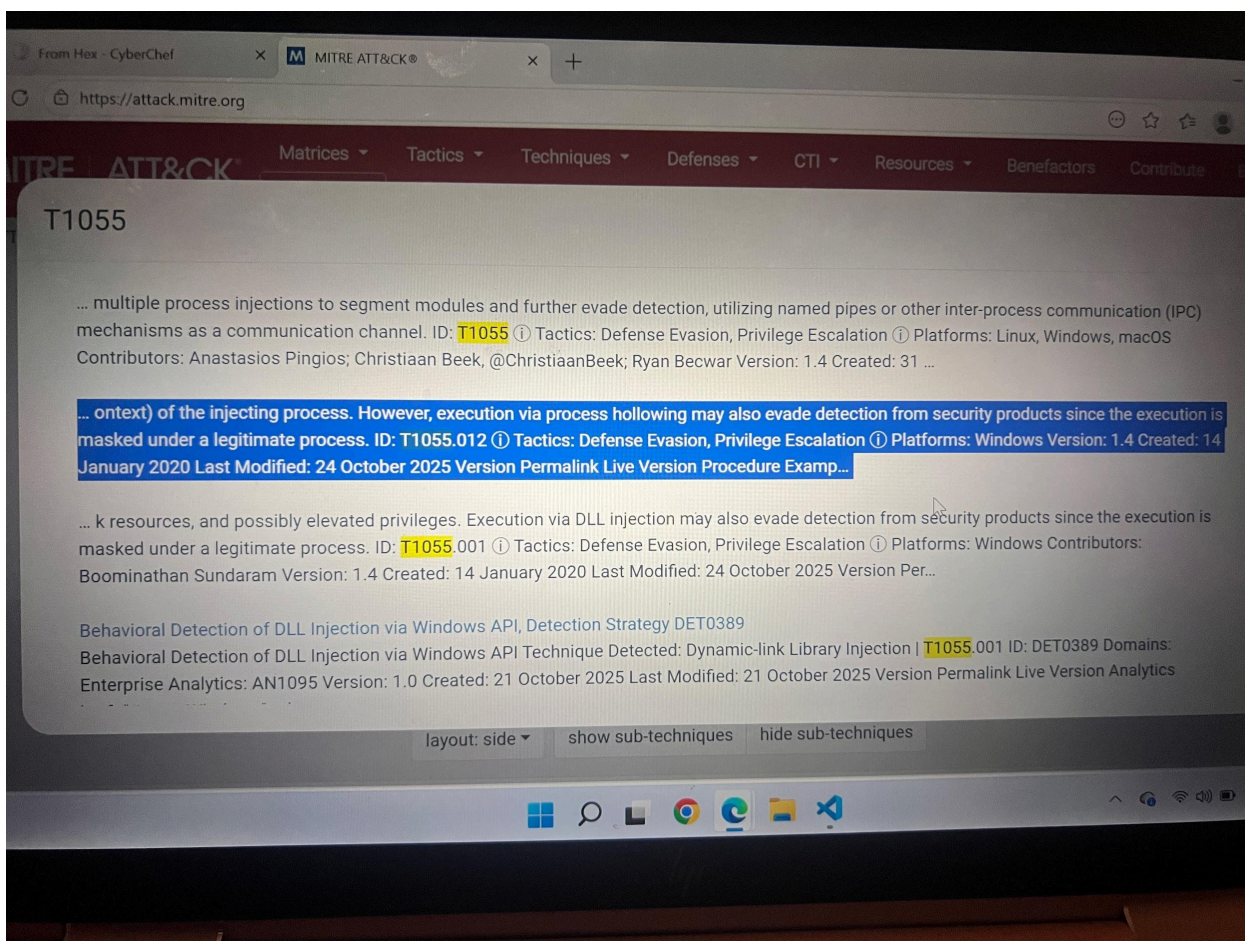
In this case, svchost.exe (PID 3880) was spawned by notepad.exe (PID 2550) and injected with a malicious executable, as evidenced by the PAGE_EXECUTE_READWRITE memory protection flag and the MZ header (4d 5a) found at memory address 0x400000.

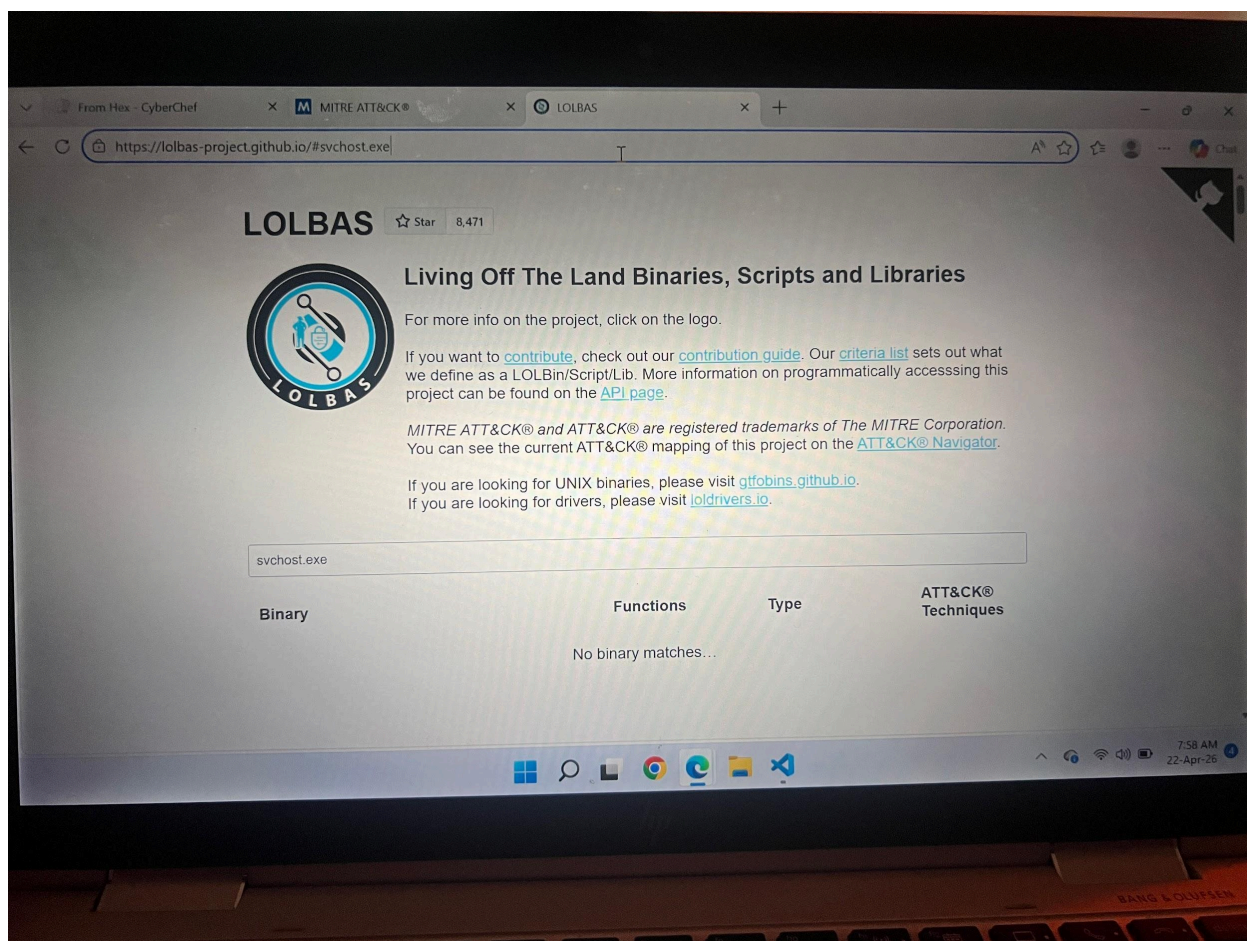
The injected process then established an outbound connection to the Command and Control (C2) server at 185.112.55.20 over port 443.

This technique is classified under the Defense Evasion and Privilege Escalation tactics in the MITRE ATT&CK Framework, which allows the attacker to execute malicious code under the disguise of a legitimate Windows process, thereby evading antivirus detection; which explains why no malicious files were found on disk.

A research on LOLBAS (Living Off The Land Binaries and Scripts) was conducted to ascertain whether notepad.exe or svchost.exe were documented as known living-off-the-land binaries.

While neither process appeared in the LOLBAS database, the behavior observed where notepad.exe was abused to spawn a system process is consistent with living-off-the-land attack patterns, a process where attackers leverage legitimate system tools to carry out malicious activity and avoid detection.





Conclusion

This investigation therefore confirms that the Finance department workstation was actively compromised at the time of the memory capture.

A threat actor successfully deployed a Process Hollowing attack by exploiting notepad.exe (PID 2550) to inject malicious code into svchost.exe (PID 3880), which then established an active communication with a remote C2 server at 185.112.55.20 over port 443.

It is recommended that the affected workstation be immediately isolated from the network and that the identified C2 IP address be blocked at the firewall level pending further investigation.